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Centralization and regional development: Evidence from a political hierarchy reform to create cities in china☆

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ABSTRACT

How does centralization or decentralization in government affect regional development? This paper draws upon a quasi-natural experiment in China's political hierarchy from the 1980s to investigate the effects of centralization in a novel sub-provincial setting. In this political hierarchy reform, China created "cities" from groups of rural counties and centralized counties' political powers to the newly established city governments. Using a difference-in-differences approach, I show that centralization has positive causal effects on industrial productivity within the created cities. My analysis of industrial firm-level data reveals a reduction in the dispersion of marginal products, suggesting that centralization will help to reduce regional resource misallocation and improve aggregate productivity. A decomposition exercise shows that the gains are mainly contributed by extensive margin from entry and exit. I quantify the loss in productivity under decentralization by hypothetically reallocating inputs to equalize marginal products to the extent observed in centralization. Finally, he reform increases the concentration and specialization of production in space.

1. Introduction

The role of political institutions in the process of urbanization and regional development has long been of interest to economists and policymakers (Henderson and Becker, 2000). Existing studies focus primarily on the consequences of democratization and political favoritism (Davis and Henderson, 2003). For example, Ades and Glaeser (1995) and Henderson and Wang (2007) show that democratization helps to limit the ability of a national ruling class to concentrate resources in the national capital. Chen et al. (2017) find that political favoritism helps high-ranking cities in China's political hierarchy grow faster by paying a lower capital price. In addition to democratization and political favoritism, this paper investigates how another important dimension of political institutions, namely, centralization or decentralization of city government powers, affects urban industrial production and regional development.

In the last three decades, many countries in the developing world have implemented decentralization reforms (World Bank, 2000). However, while the degree of centralization is considered one of the most important dimensions of policy making, in the economics literature it remains an open question. Most recent work provides ambiguous theoretical predictions on its effects (Besley and Coate, 2003; Boffa et al., 2015). Empirical evidence is also mixed, and most existing papers fail to establish a causal relationship (Bardhan, 2002; Mookherjee, 2015). Moreover, most empirical papers do not identify any clear mechanisms that might make either centralization or decentralization beneficial. In general, there is no clear answer about the extent to which a government should be centralized.

This paper aims to help fill these gaps in the literature by evaluating the causal effects of centralization stemming from a political hierarchy reform in China on regional development and quantitatively estimating the productivity gain. This hierarchy reform, first launched in 1983

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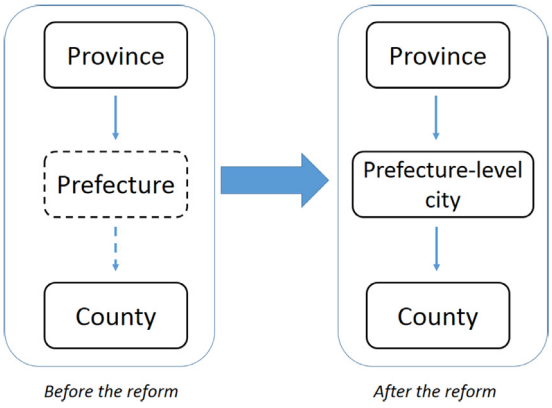


Fig. 1. Sketch of the centralization reform. *Note:* This figure plots a simple illustration of the 1983 centralization reform in China. The dashed box denotes prefectures that have no powers on component counties; the solid box denotes powers of counties are centralized to the prefecture-level city.

and lasting until 2003, created centralized "cities" to replace decentralized prefectures. The aim of the reform was to speed up urbanization through more powerful city governments. Fig. 1 provides a simple illustration of the reform. Before it took place, county governments in a prefecture could make decisions on economic projects, public finance and personnel administration in their own counties. The central government realized the market fragmentation brought by such excessive decentralization and created prefecture-level cities to take the decision-making powers from the hands of counties. Authority in county governance was then centralized to the prefecture-level cities. In a word, the creation of the prefecture-level cities was designed to centralize the powers from the counties to the cities and accelerate development of entire cities. I exploit variations in centralization induced by this reform to examine its outcomes for regional development, mainly those outcomes related to urban industrialization.

I use the variations in the reform timings of different prefectures becoming cities to conduct a difference-in-differences (DID) regression as the baseline empirical model. Using prefecture-level data from 1983 to 2003, from the baseline regression I find that, after the reform, cities experienced increases in urban industrial productivity, measured in industrial output per urban population. The primary challenge of the DID strategy is the possibility that cities were not randomly selected into the reform. The central government did state requirements on the levels of non-agricultural population and industrial output in capital counties as selection criteria. Therefore, as it focuses on how the deviations from the level relate to changes in policy, DID is a reasonable strategy. To alleviate the selection issue, in the baseline regression, I follow Li et al. (2016) to control for the trends of the key criteria. Moreover, to formally check the identification assumptions, I follow an event study approach to confirm the lack of differential pre-treatment trends across cities. Another potential threat to the identification is spillover effects between treatment and control groups. I adopt the specification in Miguel and Kremer (2004) and find no evidence of the existence of spillovers.

In addition to the baseline findings on city-level industrial productivity, I use micro data of industrial firms to shed light on the mechanism. I apply the methodology of Hsieh and Klenow (2009) to measure resource misallocation using dispersions of "revenue productivity" (TFPR) and find that the centralization reform could reduce misallocation within cities. Then, I quantify the loss in aggregate productivity due to decentralization in a counterfactual exercise. This approach demonstrates the mechanism through which centralization may be conducive to economic performance. To further determine the source of the productivity gains, I perform a decomposition exercise to distinguish the benefits from existing firms or net-entry firms. The results show that the extensive margin contributes the most. I finally document the spatial pattern of this resource reallocation process. I calculate the γ index developed by Ellison and Glaeser (1997) to measure spatial concentration of industrial production. The results show that the production in the industrial sectors became more concentrated in fewer counties.

This paper speaks to a broad literature. First, it provides novel insight into the debate between centralization and decentralization by showing that excessive decentralization is harmful and proposing a novel explanation: resource misallocation. Beyond the classical explanation that decentralization can cause externalities (Oates, 1972), recent work has placed greater emphasis on the political process. For example, while decentralization can avoid conflicts of interest (Besley and Coate, 2003) and the accountability problem (Seabright, 1996), it may induce a race-to-the-bottom competition between local governments (Keen and Marchand, 1997) and corrode state capacity by locally shielding firms from central regulations and tax collectors (Cai and Treisman, 2004). Some recent empirical studies in the context of China find that it will be associated with some loss in the environment and the safety of local workers (Jia and Nie, 2017; Shi and Xi, 2018). In this paper, I argue that decentralization can bring resource misallocation, a mechanism that has not yet been formally studied. In recent years, increasing numbers of papers have emphasized the great role of resource misallocation in explaining the disparities in aggregate productivity across countries (Restuccia and Rogerson, 2008; Hsieh and Klenow, 2009; Bartelsman et al., 2013), but few of them have quantified the contribution made due to different reasons of misallocation. In a recent study, Wu (2018) attributes capital misallocation in China to policy distortion rather than financial friction. The present paper explicitly illustrates the extent to which decentralization causes misallocation and reduces aggregate productivity.

In the specific context of China, decentralization from central to local governments is considered one of the main sources of economic growth (Xu, 2011), together with promotion incentives (Li and Zhou, 2005). Che et al. (2017) further studies the interaction between promotion incentives and the decentralization. However, Young (2000) argues against decentralization: local governments have incentives to distort the economy for rent-seeking by subsidizing specific industries, expropriating lands for industrial use or setting trade barriers. As a result, decentralization fragments the domestic market and leads production away from patterns of comparative advantage. Young (2000) observes two simple trends as suggestive evidence of the theory: the convergence in compositions of production across provinces and the divergence in prices, labor productivity and factor allocations. Poncet (2003), Bai et al. (2004) and Cai and Treisman (2006) have further explored the caveats of decentralization in China. However, to the best of my knowledge, no paper provides causal evidence on it. This paper offers rigorous empirical investigations for a solid causal relationship. In a related recent paper, Bai (2017) uses an export VAT rebate policy reform to identify the impact of internal trade barriers on TFP and misallocation.

Furthermore, my paper features a sub-provincial centralization experiment to enrich the existing literature. First, due to data availability, the previous literature mostly concentrates on "central - provincial" relations, referred to as "fiscal federalism" (Qian and Weingast, 1997). For example, since 1978, China's economic miracle has been widely attributed by numerous papers to decentralization from the central government to provincial governments (Maskin et al., 2000; Jin et al., 2005). Unlike them, I focus on the evolution of political powers between two sub-provincial governments. They represent governance entities in China but are left as black boxes in the literature without theoretical or empirical analysis. As far as I know, Zhuravskaya (2000) is the only paper with a similar "regional - local" setting, studying public goods provision in Russia.

This paper also contributes to the literature on the role of political institutions in urbanization and urban concentration. Urban economists have confirmed the importance of political institutions in determining urbanization and urban concentration (Ades and Glaeser, 1995; Davis and Henderson, 2003; Desmet and Henderson, 2015). Due to the signif-

icant role the government plays in the Chinese economy, scholars have offered empirical evidence in the context of China (Chen et al., 2017). In particular, the central government has implemented more "creating cities" experiments apart from the 1983 reform (Chung and Lam, 2004). For example, Fan et al. (2012) find that an upgrading-county reform to create cities actually fails to benefit counties' economic performance. This paper reveals that a successful policy experiment to accelerate urbanization and industrialization should involve a better resource allocation within regions.

The paper shows that a centralized political institution helps to improve resource allocation and increase the concentration of production in space. Therefore, it helps to agglomerate people in central cities and shape a core-periphery structure within a region (Krugman, 1991). These findings then speak to the literature on geographic distribution of economic activities in China. Existing studies emphasize that transportation networks contribute to shape core cities and hinterlands in China (Faber, 2014; Qin, 2017; Baum-Snow et al., 2017, 2018). Together with them, this paper provides a consistent solution to the undersized city issue in China raised by Au and Henderson (2006): eliminating restrictions on migration and internal trade for better resource allocation.

The remainder of this paper is organized as follows. Section 2 documents key features in the context of China relevant to my paper. Section 3 introduces the data. Section 4 introduces the basic empirical strategy used in the paper. Section 5 presents the empirical results. Section 6 concludes.

2. Background

2.1. Political hierarchy and jurisdictions in china after 1978

In China, there are four levels in sub-national jurisdictions: province, prefecture, county and township level. Appendix Figure A.1 provides some basic information. The average size of a province in China is approximately 300,000 km^2, similar to the sizes of Italy or Arizona. As a result, it is difficult for provincial governments to make policies according to local heterogeneity. They must rely on lower level governments, that is, prefecture- and county-level governments, to implement policies locally. Counties are the basic units in China's local government hierarchy, comparable to counties in the United States. Nevertheless, it is not easy to find a comparable counterpart to prefecture-level jurisdictions in the United States. We can use Metropolitan Statistical Areas (MSA) to form a rough analogy, though MSAs are not legal administrative units. Similar to an MSA, in China a prefecture-level jurisdiction combines several counties. Its average size is approximately 28,000 km^2, similar to the size of the Chicago metropolitan area, a combination of fourteen counties. In terms of population, its average population is approximately 4.1 million by 2010, similar to the San Francisco metropolitan area.

As prefecture- and county-level governments play an essential part in China's political and economic processes, how governance responsibilities and powers are divided is critical to understanding local governments in China. This paper concentrates on the relations between prefecture- and county-level governments. Fig. 1 sketches the basic political hierarchy in China. A province is composed of several prefecture-level units, and a prefecture-level unit is composed of several county-level units. Prefecture-level governments play an intermediate role between a province and its component counties. There are two types of prefecture-level units: prefecture (*diqu*) or prefecture-level city (*diji shi*,). It must be emphasized that both prefectures and prefecture-level cities are at the same level (prefecture level) in China's political hierarchy. My analysis is based on the transition from prefectures to prefecture-level cities at this same level.

A key difference between a prefecture and a prefecture-level city[1] lies in their relations with the counties belonging to them: in a prefecture, decision powers, including personnel, fiscal and administrative, are decentralized to county-level governments, and in a prefecture-level city these powers are centralized to the prefecture-level government.

In a prefecture, all counties are nearly autonomous, and the county governments have decision rights over the development of their own counties. The prefecture government possesses no power over the counties, since in legal terms it is only a provincial government's "resident agency" (*paichu jigou*) in its prefecture's territory. In other words, it does not constitute an official layer of local administration (Chung and Lam, 2004). Leaders in the prefecture government cannot enact economic policies on their own; instead, they convey decisions by the provincial government to their component counties and oversee them. Therefore, counties under any prefecture can make independent decisions on their own affairs, including those of an economic and political nature. For example, they can make decisions about where public funds shall be spent and what public goods will be offered, about local taxes and subsidies, and about the location of newly opened state-owned enterprises, among other areas. In general, in a prefecture, counties enjoy considerable autonomy.

In contrast, in a prefecture-level city, county governments lose their decision rights. The prefecture-level city government located in the capital county can administer the development of all of the prefecture-level city's counties. The reason for this difference is that the prefecture-level city government is not a provincial government's "resident agency" as in the prefecture case but rather an essential intermediate level of government between a province and a county. Legally, its component counties are under the supervision of the prefecture-level city government itself, not of the province government. In this situation, county governments are manipulated by the prefecture-level city government. Their leaders are nominated by the prefecture-level city government, and their fiscal revenue and expenditures are not only supervised but also managed. They are not allowed to set up new state-owned enterprises freely and the choice of location is coordinated by the prefecture-level city government. Counties in prefecture-level cities are therefore not as autonomous as in prefectures. City officials can give a range of orders to any given county.

In general, three types of decision rights are transferred to city governments (Shi et al., 2009). First come administrative powers with regard to social and economic projects. A prefecture-level city government has hundreds of administrative rights over developments in its constituent counties. For example, the establishment of a state-owned enterprise can no longer be decided by the county in which it is located. Since the reform, this decision must be approved by the corresponding prefecture-level city government. Second are powers over public finance revenue and expenditure. Prefecture-level cities are a formal independent fiscal regime, while prefectures are not. On the revenue side, prefecture-level city governments now take part in sharing counties' tax revenue and the counties must rely more on transfer payments from upper levels of government. On the expenditure side, prefecture-level city governments take more responsibility for public expenditures. Third come the powers to appoint, reward and punish local officers. County officers must therefore obey orders from prefecture-level cities. These powers in three areas of decentralization are similar to those summarized by Bardhan (2002) and World Bank (2000): administrative, fiscal and personnel. The combination of these powers is important. For example, Blanchard and Shleifer (2001) show that centralization of political powers on personnel is a key complement to fiscal federalism.

An example documented by a local officer in Wuhu, Anhui Province provides a vivid illustration on the function of prefecture-level city governments. In his book, Han (1986), describes how Wuhu prefecture-level city government managed its component counties: "Due to huge duck down resources in Wuhu, every county wanted to build a down coat factory. But the prefecture-level city government only approved one factory in the capital county to avoid waste, which could not happen under the previous prefecture setting." We can see from this story that a prefecture-level city government can manage public finance funding

[1] I use "city" occasionally in the paper for short.

and decide on several economic projects on component counties, which is impossible for prefecture governments.

It is noteworthy to distinguish two types of component counties in prefectures or prefecture-level cities: capital counties (or urban districts/core cities, *shiqu*) and peripheral counties (or rural counties, *xian*). A capital county is where the prefecture or prefecture-level city government is located. Historically, a capital county was the central and the most urbanized part in a prefecture or prefecture-level city, so it is called urban districts or core cities in other papers. Its identity as the location of the prefecture-level government is mostly confirmed in history and remains unchanged. In contrast, as their name suggests, peripheral counties normally surround the capital counties with a lower level of urbanization and development. The predetermined differences between capital and peripheral counties allows the present study to explore possible distributional effects on component counties.

2.2. The 1983 reform: "turning prefectures into cities"

The reform starting in 1983 aimed to abolish prefectures and subject counties to the rule of prefecture-level cities. In other words, it collected groups of counties and created prefecture-level cities as the center of the counties. The primary goal of the reform was to accelerate urbanization and industrialization in entire cities through better coordination of resources across the component counties: "The shortcomings of the prefecture institution is obvious: within a region, too many local governments exist; their works and policies contradict and offset each other, producing fragmentation; and it is harmful for social and economic coordination and development ⋯ The main solution is to gradually transform prefectures into prefecture-level cities and let prefecture-level cities lead component counties" (Central Committee of the Party and the State Council, 1983). The central government supposed that after a prefecture had been changed into a city, the city government could better manage and coordinate component counties. The central government expected this prefecture-level city setting to increase the efficiency of component counties, including both capital and peripheral, under the coordination of city governments.

The timing of the reform was mainly made by the central government (Chung and Lam, 2004). In Central Committee of the Party and the State Council (1983), the central government emphasized that the provincial governments should not rush the reform and that all reform proposals should be approved by the central government. It also clarified the basic guidelines about this reform. Among them, two criteria were to be met on which kind of prefectures could be reformed. The first required the non-agricultural population in capital counties to be 150,000. The second required the industrial output in capital counties to be 400 million yuan. The central government believed that those cities should "play a larger role in managing counties". It is noteworthy that these criteria only required levels rather than growth of key indicators. Furthermore, the indicators were scaled economic statistics (population and output) rather than productivity-based statistics. Nevertheless, these conditions were not binding in the actual process. The process of this reform is illustrated in Fig. 2. Starting from 1983, the number of prefecture-level cities increased rapidly, and the number of prefectures decreased at a similar rate. Until 2003, apart from a few special minority residences, almost all prefectures were turned into cities. Moreover, as the pattern of the number of total prefecture level jurisdictions suggests, such a reform was basically a one-on-one transition between a city and a prefecture.

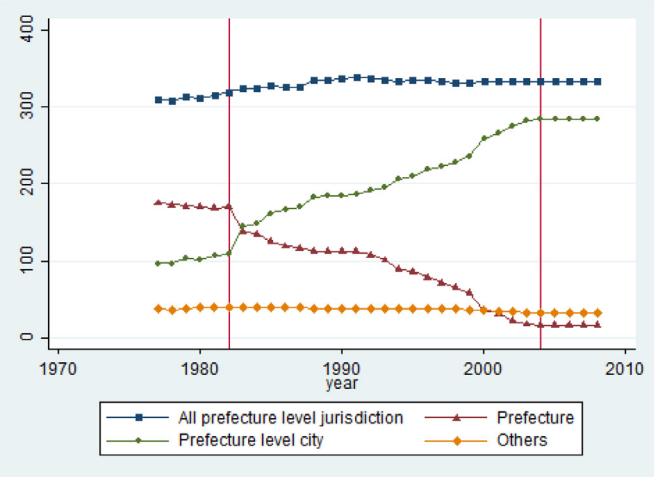


Fig. 2. Changes in the number of prefecture level jurisdictions.
Note: This figure plots the numbers of various types of prefecture-level jurisdictions in China over the years.

3. Data

3.1. Aggregate data

The main outcomes of interest relate to the industrial and urban development in China. To this end, I collect and digitize data on output and population both on a prefecture level and a county level. Output variables can be broken into agricultural and industrial. Population variables can be broken into agricultural and non-agricultural. They are reported by the National Bureau of Statistics in published statistical yearbooks and local gazetteers.

The study period in this paper is from 1983 to 2003. Prior to 1983, data are scarce; after 2003, the central government stopped the reform and began to re-decentralize powers to some developed counties under various prefecture-level cities, which may contaminate the estimated effects of the centralization reform. Li et al. (2016) show that another reform since 2003, aiming to remove the power of cities, had a negative impact on the local economy, which is consistent with the findings in this paper.

In the empirical analysis, I do not include five provinces: Tibet, Inner Mongolia, Xinjiang, Qinghai and Hainan, since their prefectures did not undergo reform during the time period examined, so there are no within-province variations. The reason for this is low population densities and certain special policies in the first three of them (e.g., the ethnic autonomous institution). Furthermore, I exclude those that had already been prefecture-level cities before 1983. As they are not a part of the 1983 reform, they may not be comparable to those treated during the reform (Chung and Lam, 2004). Most of them are capitals of provinces and much larger than those in the sample. Finally, I exclude four province-level cities (Beijing, Shanghai, Tianjin and Chongqing) as well as Hong Kong, Macau and Taiwan, for which the reform did not apply. In the end, the policy affected approximately 60% of the population in China as a whole.

Panel A of Appendix Table A.1 provides summary statistics of the population and output aggregates. We can see that capital counties were more urbanized and industrialized than peripheral counties and even showed higher productivity in industrial production. Their output made up the majority of output in the prefectures.

Apart from the outcomes of population and production, the variables associated with public finance are also significant for confirming that the centralization actually occurred. The public finance data come from the statistics reported by China's Ministry of Finance. I digitize fiscal revenue and expenditures at both the county and prefecture level annually. Summary statistics of the main public finance variables are presented in Panel B of Appendix Table A.1.

3.2. Firm-level statistics

To explore more implications of the effects on industrial sectors and on enterprise, I use firm-level data from China's Annual Survey

of Industrial Production, a survey also conducted by the National Bureau of Statistics. It is an annual census from 1998 containing all non-state industrial firms with sales more than 5 million yuan (approximately 0.6 million dollars in 1998), plus all state-owned firms. This survey represents the most widely used micro data when studying industrial production in China (for example, Hsieh and Klenow (2009) and Song et al. (2011)).

To fit into the study period of the aggregate level analysis, I use firm-level data from 1998 to 2003. In that period, 55 prefectures received the reform, compared with 98 treated between 1983 and 1997. Therefore, even though it covers a short period, the firm-level dataset provides enough variations for empirical investigations. I use unique IDs to link firms over years to construct the panel. In some special cases, such as mergers and acquisitions, firms change their IDs. To provide more precise matching, I follow Brandt et al. (2012) to link firms over time using their Chinese name, address and telephone number in addition to unique IDs.

The information I use from this firm-level dataset includes the firm's industry code, location, ownership, outputs, value-added, wage payments and capital stock.

4. Empirical strategy

4.1. Baseline model

To empirically evaluate the impact of the reform, I use a difference-in-differences (DID) method as the baseline model:

$$Y_{jpt} = \beta Treat_{jpt} + \phi w_{jpt} + \alpha_j + \delta_{pt} + \epsilon_{jpt} \tag{4.1}$$

where Y_{jpt} is an outcome variable in prefecture j, province p and year t; $Treat_{jpt}$ is the main independent variable, indicating whether prefecture j in province p received treatment by year t to become a prefecture-level city; α_j is a prefecture fixed effect; δ_{pt} is a year × province fixed effect; w_{jpt} are control variables, first including some simultaneous administrative policy changes: for example, whether prefecture j becomes a special economic zone, a coastal open city or a deputy-provincial city (Wang, 2013). I cluster the standard errors at the province level. Thus, β provides the overall effects of the centralization reform on outcomes of interest in prefectures as a whole.

The inclusion of prefecture fixed effects α_j captures any time invariant characteristics of prefectures/cities, such as culture and geography. Province × year fixed effects δ_{pt} pick up province specific shocks, for example, policies implemented by provinces or the central government, price fluctuations and changes in central-local relations. Controlling for these fixed effects, I identify the effects of the centralization reform using within-province variations.

Variations of the main independent variable $Treat_{jpt}$ come from different reform timings in the sample. The reform exhibited considerable variation in the extent of centralization. Fig. 3 depicts the variations of the reform across space. The spatial variation of reform timing raises a natural challenge to the validity of the empirical strategy, i.e., the non-randomness of the reforms across prefectures. As introduced in the background section, the central government stated two criteria used to select cities into the reform, the levels of non-agricultural population and industrial output. Two features of the criteria justify the rightness of DID as the identification strategy. First, the criteria are based on the levels of indicator rather than growth, while the DID specification uses fixed effects or first-difference to focus on growth. Second, the criteria concern scale indicators such as population and output, whereas this paper will focus on the productivity measures.

Furthermore, I add more control variables to address the potential challenge to the identification apart from the baseline policy controls. I follow Li et al. (2016) to control for the interactions between the two key selection criteria (non-agricultural population and industrial output in capital counties) in the initial year and a series of year dummies. It will help to address differential trends in outcomes between treatment and control groups caused by the criteria. I also control for interactions between several geographic conditions and year dummies, including a prefecture's distance from the nearest river and coast line and the corresponding provincial capital city. Although in the baseline DID specification, all the time-invariant impacts brought by heterogeneous geographic conditions as well as initial levels of the selection criteria are absorbed by prefecture fixed effects, controlling for their potential time-variant effects is still helpful to address the non-randomness of the reform. Finally, I will add prefecture-specific time trends to the baseline regression. The individual-specific time trends in a fixed-effect model will allow treatment and control groups to follow different trends. If the estimated coefficients do not change considerably, it will greatly support the identification strategy of the baseline model.

4.2. Checks on the identification assumption

Though controlling for a large number of observables and fixed effects, it is still possible that there were some unobservable factors that were simultaneously correlated with reform timing and outcomes, which would bias the estimation. For example, it could be the earlier selection of those prefectures with more growth potential as prefecture-level cities. A formal test of the identification assumption is required. As in any DID specification, the identification assumption of the baseline regression is a standard parallel trend assumption: in the absence of reform, the growth in the outcomes of interest would be the same across any prefectures within a province, conditional on all observables.

To show that there are no differential trends prior to the reform, I propose a flexible DID model to show the trends of the treatment effects before and after the reform year. To be specific, I test the identification assumption by estimating a set of yearly treatment effects beginning four years prior to the reform event and continuing for three years thereafter. This enables me to check pretrends in these yearly treatment effects to secure the rightness of the identification assumption. It is a more flexible form of the baseline regression to allow the effect to vary by year in relation to the reform. The specification can be written as follows:

$$Y_{jpt} = \sum_{\tau=-4}^{3} \beta_\tau I(YearsSinceTreat_{jt} = \tau) + \phi w_{jpt} + \alpha_j + \delta_{pt} + \epsilon_{jpt} \tag{4.2}$$

where $I(\cdot)$ is an indicator function and $YearsSinceTreat_{jt}$ counts the years at time t since prefecture j was treated to cities. Then, $YearsSinceTreat_{jt}$ takes negative values counting the years before the treatment, positive values after the treatment and zero when t is the year it was treated. The effects beyond +3 and -4 year are grouped into +3 and -4 respectively. I set the year just prior to the reform as the omitted group, so all the coefficients are relative to the gap in the -1 year. If the parallel trend assumption holds prior to the reform, $\beta_\tau = 0$ when $\tau < 0$.

In addition to the standard parallel trend assumption, the identification also requires no spillovers between treatment and control groups. The DID model compares the relative effects of the reform on treated and untreated prefectures. It assumes that the untreated prefectures were not affected by the reform. However, it is possible that the reform attracts firms and citizens from untreated prefectures to relocate in treated prefectures. If such spillovers exist, the implication will be a relocation effect rather than a growth effect.

To formally check the non-spillover assumption, I adopt the specification in Miguel and Kremer (2004) to estimate the spillover effect:

$$Y_{jpt} = \beta Treat_{jpt} + \lambda NeighborTreat_{jpt} + \phi w_{jpt} + \alpha_j + \delta_{pt} + \epsilon_{jpt} \tag{4.3}$$

where $NeighborTreat_{jpt}$ indicates the treatment status of prefectures neighboring to prefecture j. Therefore, λ captures the spillover effect. If no spillovers exist, we will expect $\lambda = 0$.

5. Results

In this section, I evaluate the impact on industrial activities in cities using aggregate and firm-level data. To begin, I use prefecture-level data

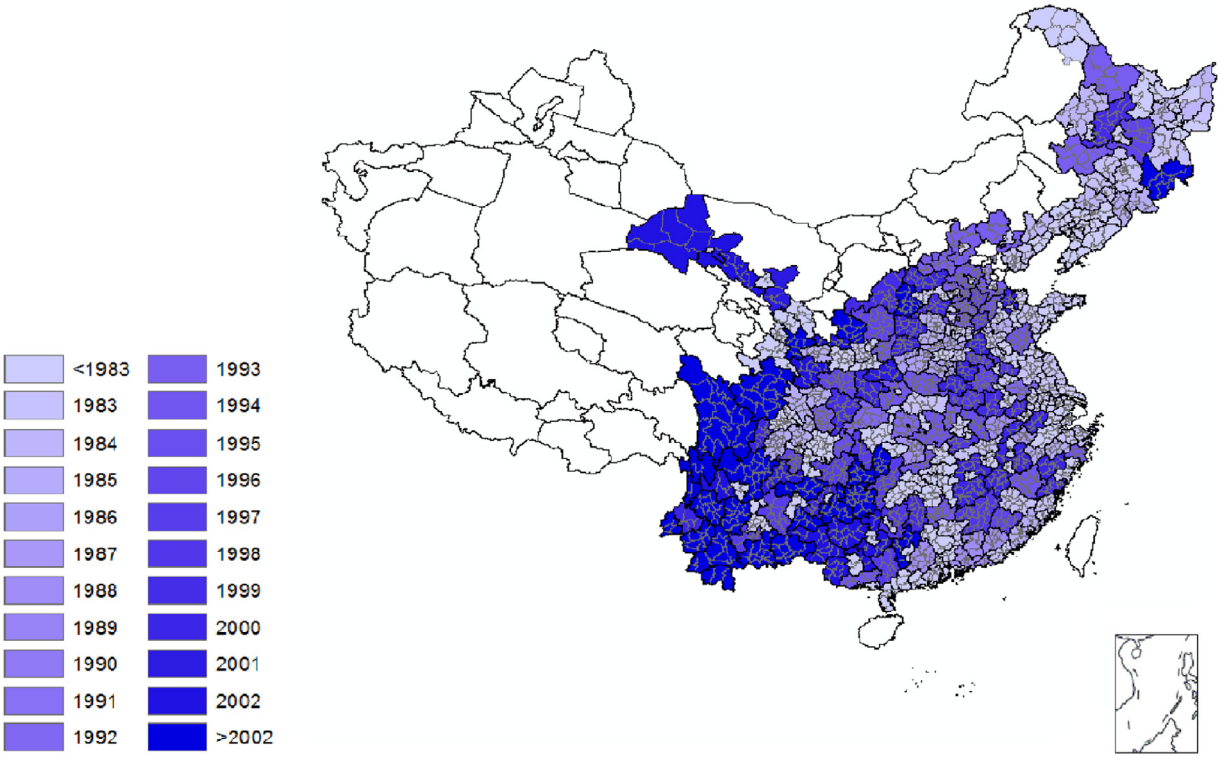


Fig. 3. Timing of the reform across space.
Note: This figure graphs the nationwide reform timing. The darker the color, the later the treatment happens. The white zones are those excluded in the sample.

on population and outputs to calculate industrial productivity and estimate the baseline DID regression. I then perform various checks on its causal interpretation, including the lack of differential pre-treatment trends and spillover effects. Then, I turn to analysis using micro data of industrial firms. I quantify reductions in resource misallocation and resulted gains in aggregate productivity as an important channel through which centralization can be beneficial. I further decompose the productivity gains and find that the net-entry firms contribute the most to the aggregate productivity gains. I next show that the centralization reform and reallocation of resources is associated with geographic concentration and specialization in industrial production within cities. Finally, I confirm that centralization did occur at the prefecture level after the 1983 reform with data on public finance, as the premise of the story. In general, the results reveal that the centralization reform benefits urban industrial activities through a better resource allocation within cities, together with sizeable gains in productivity and industrial concentration in space. The findings are in line with Young (2000)'s observation that decentralization in China induces market fragmentation and divergence in prices and productivity, which are signs of distortions in production and resource allocation.

5.1. Baseline results

In this subsection, I employ the baseline DID model (Eq. (4.1)) to show that centralization will increase prefecture-level urban industrial productivity. The main outcome of interest, urban industrial productivity, is measured in industrial output per urban population. Specifically, I calculate it using prefecture-level industrial output divided by non-agricultural population and then taking the natural logarithm. I will carry out several checks on its identification assumption to ensure that we can interpret the baseline results as causal effects in the next subsection.

Table 1 presents the baseline results. The outcome of interest is urban industrial productivity, measured by the log value of the industrial output per urban population. To realize a DID setting, in all regressions I control for prefecture and province × year fixed effects. To account for simultaneous city-based policy changes, I include three dummies indicating whether it becomes a special economic zone, a coastal open city or a deputy-provincial city. Column (1) shows that after a prefecture becomes a prefecture-level city, its increase in urban industrial productivity is 12.9% more than those remaining as prefectures. The difference is significantly different from zero at the 1 percent level. Then, I add other sets of control variables, as described in Section 4.1, grad-

Table 1
Baseline model: prefecture-level industrial productivity.

VARIABLES	(1)	(2)	(3)	(4)
	log(industrial output per urban population)			
Treat	0.129***	0.170***	0.142***	0.135**
	(0.040)	(0.052)	(0.047)	(0.051)
Prefecture FE	✓	✓	✓	✓
Year × province FE	✓	✓	✓	✓
Policy controls	✓	✓	✓	✓
Geographic conditions × year FE		✓	✓	✓
Initial conditions × year FE			✓	✓
Prefecture time trends				✓
Observations	1568	1568	1568	1568
R-squared	0.962	0.975	0.976	0.986

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variable is log(industrial output per urban population). The main independent variable is whether this prefecture is treated to a prefecture-level city. Prefecture fixed effects and year × province fixed effects are controlled in all columns. Policy controls include dummies indicating special economic zone, coastal open city and deputy-provincial city; geographic conditions include interactions between year dummies and distance to the nearest river, coast line and provincial capital city; initial conditions include interactions between year dummies and industrial output and the non-agricultural population in the initial year; prefecture time trends include interactions between linear time trends and prefecture fixed effects. Standard errors are clustered on province level.

ually into the baseline model. In column (2) I add the interactions between year dummies and the prefecture's distance from the nearest river and coast line and the corresponding provincial capital city. Column (3) further includes the interactions between year dummies and the prefecture's industrial output and non-agriculture population in the initial year, the two primary determinants in the selection into the reform. Including these interactions helps to address the non-randomness selection concern (Li et al., 2016). The magnitudes of the coefficients are similar as in the first column and remain statistically significant at the 1 percent level. In the last column, I add prefecture-specific trends in the regression to allow all prefectures to follow different linear trends in their industrial productivity. Column (4) shows an effect of 13.5% after accounting for prefecture-specific trends, significant at the 5 percent level. This result adds in the confidence of the DID model (Angrist and Pischke, 2010).

The results are reasonable in the light of the main idea of this paper. When a prefecture is transformed into a prefecture-level city, the creation of the city government will take the decision powers from counties. The centralization of governance in cities helps to improve the efficiency of urban industrial production. In that way, the reform successfully creates productive industrial cities.

The above results still leave a significant concern unmet: the possible endogeneity of the treatment. Though several sets of control variables are included and help to address endogeneity concerns, formal checks are needed to confirm the causal relationship.

5.2. Identification

The first and foremost identification assumption to test is the parallel trend assumption. It requires the growth in the outcomes of interest to be the same across any prefectures in the absence of reform. If the assumption does not hold, the above results cannot be interpreted as causal effects. For example, if prefectures were selected into the reform by their unobserved characteristics, such as growth potential, the DID strategy could fail. Huang et al. (2017) note some factors affecting the decentralization decisions by the central government. To test the assumption formally, as introduced in Section 4.2, I adopt an event study approach to estimate a flexible DID model 4.2 that allows coefficients to vary year by year. We can then observe any possible differential growth in the outcome variables and check the parallel trend assumption. Table 2 presents the results. To show it is more straightforward, Fig. 4 plots these coefficients along the relative years to the reform event, as well as the associated 95% confidence intervals.

The results in Table 2 and Fig. 4 show that there are no significant pre-existing differential trends in the growth of industrial productivity: the coefficients before the reform are all insignificant and close to zero. Thus, prefectures treated earlier or later in calendar years followed a similar trend in the outcome of interest before the reform. In other words, the absence of evidence on differential pretrends suggests that the central or provincial level government did not choose the targets of the reform by the growth rate of the non-agricultural population or industrial output. Therefore, the positive effects in the baseline regressions, reported in Table 1, can be interpreted causally with confidence.

In addition to the parallel trend assumption, the other identification assumption for the DID strategy requires that there are no spillovers between treatment and control groups. The possible presence of the spillovers indicates that the treatment effects in the reformed cities can just be a relocation of industrial activities from neighboring prefectures. To formally check the spillover effects, I estimate Eq. (4.3), proposed by Miguel and Kremer (2004). The coefficients of treatment status of neighboring prefectures capture the spillover effect. In the first column of Table 3, I estimate the possible spillovers from contiguous neighbor prefectures; in the second column, I further include prefectures that are not contiguous but that lie in the same province. Table 3 shows that there is no evidence of spillovers from both types of neighboring prefectures. It suggests that the positive effects in the baseline regressions are not due to a relocation or other spillover from control groups.

The above combination of the lack of differential pretrends and spillover effects addresses the concerns about the identification issue of the treatment variable. After these checks on the identification assumption, it can be credibly stated that the treatment effects estimated in the baseline regressions reflect a causal relationship between the creation of cities and industrial development.

Table 2
Identification: event study results.

VARIABLES	(1) log(industrial output per urban population)
Treat (−4)	−0.044
	(0.040)
Treat (−3)	−0.052
	(0.063)
Treat (−2)	−0.001
	(0.055)
Treat (0)	0.158***
	(0.045)
Treat (+1)	0.141***
	(0.047)
Treat (+2)	0.146**
	(0.056)
Treat (+3)	0.191***
	(0.054)
Prefecture FE	✓
Year × province FE	✓
Controls	✓
Observations	1568
R-squared	0.975

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variable is log(industrial output per urban population). The main independent variables are a series of dummy variables indicating whether the year of observation falls in a given relative year as measured from the year of the reform happened. Prefecture fixed effects and year × province fixed effects are controlled in all columns. Controls include all control variables in the baseline model. Standard errors are clustered on province level.

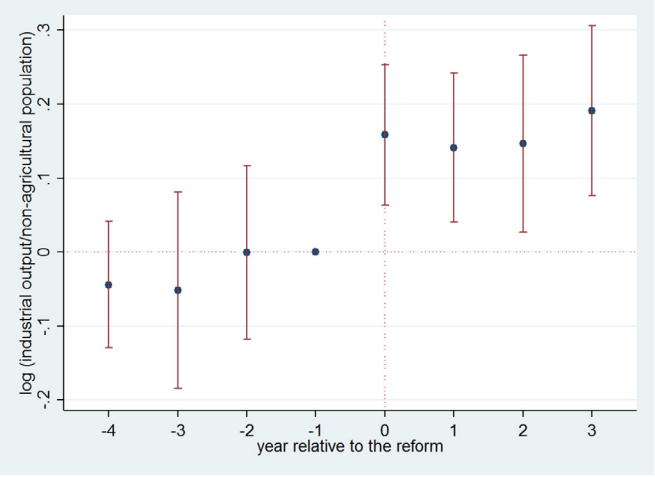


Fig. 4. Event study graph.
Note: This figure reports estimates of the dynamic effect of the reform derived from a flexible DID specification. Estimates are constructed by regressing the outcome variable on a series of dummy variables indicating whether the year of observation falls in a given relative year as measured from the year the reform took place. Relative year -1 is the omitted category so that all estimates should be interpreted as relative to the 1st year prior to the reform. The vertical solid lines plot the 95 percent confidence interval for the estimates.

Table 3
Identification: spillover effects.

VARIABLES	(1)	(2)
	log(industrial output per urban population)	
Treat	0.136***	0.163***
	(0.043)	(0.054)
Contagious treat	−0.033	0.001
	(0.026)	(0.039)
Same-province treat		−0.073
		(0.059)
Prefecture FE	✓	✓
Year × province FE	✓	✓
Controls	✓	✓
Observations	1568	1568
R-squared	0.985	0.985

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variable is log(industrial output per urban population). The main independent variables are whether this prefecture is treated, and whether neighbor prefectures are treated. Prefecture fixed effects and year × province fixed effects are controlled in all columns. Controls include all control variables in the baseline model. Standard errors are clustered on province level.

5.3. Misallocation and aggregate productivity

The baseline results provide empirical evidence on the causal relationship between centralization and urban industrial productivity measured at the city level. The results imply that creating cities should be associated with increases in production efficiency. Centralization can bring better coordination and allocation of resources, which will lead to an improvement in aggregate productivity. The reduced misallocation and the increases in aggregate productivity are the main ways in which this paper addresses the benefit of centralization. In this subsection, I will provide more precise evidence using TFP derived from firm-level data to identify the existence of this mechanism.

The problem of resource misallocation in developing countries has attracted growing attention in recent years. Hsieh and Klenow (2009) provides a methodology for evaluating the contribution of misallocation in explaining low aggregate output per worker and total factor productivity (TFP) in developing countries. In this paper, the authors document the dispersion of "revenue productivity" (TFPR) as a proxy of resource misallocation, and then measure how much aggregate manufacturing output in China and India could benefit if the marginal products of labor and capital were equalized to the extent observed in the United States. They also show briefly that the extent of misallocation can be affected by varied policy distortions such as licensing and size restrictions. More recent literature (Bartelsman et al., 2013; Midrigan and Xu, 2014) provides additional evidence on the role of misallocation in explaining the TFP gap in a similar framework.

However, when calculating how misallocation can account for the TFP gap, the previous literature has placed less emphasis on its sources and possible remedies, although they all agree that institutions and policies may be responsible for distortions and misallocation (see a survey by Hopenhayn (2014)). In this part, I investigate whether the problem of misallocation is mitigated by more centralized institutions. Intuitively, the decentralized institutions before the reform would seem to encourage county governments to subsidize and protect local enterprises. This would have resulted in distortions in efficient resource allocation within a prefecture. Young (2000) attributed the divergences in price and productivity, a sign of resource misallocation, to the excessive decentralization to local governments in China. If decentralization is responsible for misallocation, as previous studies have predicted, we should expect that after a centralizing reform, the prefecture-level city government reduces misallocation by reallocating resources and then promoting efficiency in the whole region.

To measure the misallocation, I borrow the methodology of Hsieh and Klenow (2009) to calculate dispersions of "revenue productivity" (TFPR) as a measurement. I assume each firm i in sector s has a diminishing returns production function:

$$Y_{is} = A_{is}\left(K_{is}^{\alpha_s} L_{is}^{1-\alpha_s}\right)^{\eta}, \tag{5.1}$$

where A_{is} is the total factor productivity of the firm u, α_s is the share of capital in sector s and η is the degree of decreasing returns. The final product is a Cobb-Douglas aggregate of products in different sectors. Next, I define the TFPR of each firm i in sector s as Hsieh and Klenow (2009):

$$TFPR_{is} = \frac{P_s Y_{is}}{K_{is}^{\alpha_s} L_{is}^{1-\alpha_s}} \tag{5.2}$$

The essence of TFPR can be regarded as a measurement of marginal products or distortions faced by firms:

$$TFPR_{is} \propto MRPK_{is}^{\alpha_s} MRPL_{is}^{1-\alpha_s} \propto (1+\tau_{is})^{-1} \tag{5.3}$$

Thus, a high firm TFPR implies that this firm faces a relatively low subsidy or barriers that raise its marginal products, suggesting the firm is smaller than its optimum size. To see how the dispersion of TFPR can proxy for resource misallocation and be responsible for a loss in aggregate TFP, I write the aggregate TFP in prefecture j and sector s as within the monopolistic competition framework:

$$TFP_{js} = \left[\sum_i \left(A_{is}\frac{\overline{TFPR_{js}}}{TFPR_{ijs}}\right)^{\frac{1}{1-\eta}}\right]^{1-\eta} \Big/ (K^{\alpha_s} L^{1-\alpha_s})^{\eta} \tag{5.4}$$

Following from Jensen's inequality, if and only if marginal products are equalized across firms, or if there is no dispersion in distortion rate or TFPR, aggregate TFP can reach its optimum.

I calculate TFPR following Hsieh and Klenow (2009), who also use China's Annual Survey of Industrial Production. The capital share α_s in each sector s is set to be the same in the corresponding industry in the United States, which is taken from the NBER Productivity Database. I trim the 1% tails of distribution of TFPR across industries. I follow Hopenhayn (2014) to set the degree of decreasing returns $\eta = 0.85$[2]

Under centralization, the city government can reallocate funds across counties and help to reduce the extent of dispersion of distortion rates. From Eq. (5.3), we can see when the distortion rates τ_{us} faced by different firms become less dispersed, dispersions of TFPR will also be reduced. Therefore, the aggregate TFP within a prefecture will improve according to Eq. (5.4).

To illustrate that centralization can reduce misallocation, as an example in Appendix Figure A.2 I plot distributions of TFPR in Ningde when it was a prefecture and when it was reformed to a city. We can see that after the centralization reform, the distribution of TFPR in Ningde is clearly less dispersed.

As a formal analysis, I propose two different methods to show how misallocation and aggregate TFP are associated with the status of prefectures or cities.

To begin, I use my baseline regression model to check any potential changes in misallocation associated with the city-creating reform:

$$Misallocate_{jst} = \beta Treat_{jt} + \phi w_{jpt} + \alpha_j + \rho_{st} + \delta_{pt} + \epsilon_{jst} \tag{5.5}$$

where $Misallocate_{jst}$ is the dispersion in TFPR within prefecture j, sector s and year t; w_{jpt} are baseline controls; α_j is prefecture fixed effects; ρ_{st} is year × sector fixed effects; δ_{pt} is year × province fixed effects. Here, I use standard deviations, ratios of the 75th to 25th percentiles and ratios of the 90th to 10th percentiles in TFPR within sectors and prefectures as measurements of the dispersion in TFPR, and all of them are standardized, as in Hsieh and Klenow (2009). The regression results are in Panel A of Table 4.

[2] Hsieh and Klenow (2009) assumes the diminishing returns occur on the demand side with the monopolistic competition model, isomorphic with my model. Their choice of parameters implies $\eta = 0.5$, a relatively low value for diminishing returns. Hopenhayn (2014) suggests $\eta = 0.85$ as a more common choice. I check robustness for $\eta = 0.5$ in appendix.

Table 4
Results on Misallocation.

VARIABLES	(1) TFPR SD	(2) TFPR 75-25	(3) TFPR 90-10
Panel A: Misallocation at prefecture level			
Treat	−0.127***	−0.093**	−0.134***
	(0.042)	(0.044)	(0.034)
Prefecture FE	✓	✓	✓
Year × province FE	✓	✓	✓
Year × sector FE	✓	✓	✓
Observations	16,900	16,900	16,900
R-squared	0.082	0.043	0.055
Panel B: Misallocation at county level			
treat	−0.005	−0.012	−0.027
	(0.054)	(0.026)	(0.023)
County FE	✓	✓	✓
Year × province FE	✓	✓	✓
Year × sector FE	✓	✓	✓
Observations	81,215	81,215	81,215
R-squared	0.075	0.018	0.023

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variables are measurements on dispersions of TFPR. The main independent variable is whether this prefecture is treated to a prefecture-level city. Prefecture fixed effects, year × province and year × sector fixed effects are controlled. Standard errors are clustered on province level.

From the results we find that all three measurements of the dispersion of TFPR are significantly reduced after the reform. The results suggest that the centralization reform can mitigate resource misallocation within a prefecture, as my story suggests.

I repeat the same practice on a county-sector level regression by aggregating dispersions of TFPR within each county-sector pair instead of prefecture-sector pair. Intuition shows that the reform in the prefecture level can improve the allocation across counties but not within counties. Thus, coefficients of the treatment variable in county-sector-level regressions are not expected to be significantly negative, as shown in Panel B of Table 4. Looking across column (1) to column (3), we cannot see any significantly negative effects on the dispersion of TFPR within counties. No stable and robust relationship can be found between the centralization reform and misallocation in the within-county analysis. This regression can be regarded as a "placebo" as opposed to the actual effect on misallocation across counties. Such results tend to confirm that the centralization reform improved allocation efficiency but that this was not due to other spurious correlations to reduce misallocation measurements.

Second, I conduct a counterfactual analysis similar to Hsieh and Klenow (2009). With Eq. (5.4) in mind, I can calculate the efficient aggregate TFP. In that case, marginal products are equalized across firms in a given sector within a prefecture:

$$TFP_{js,EFF} = \left[\sum_i (A_{is})^{\frac{1}{1-\eta}}\right]^{1-\eta} / (K^{\alpha_s} L^{1-\alpha_s})^{\eta} \tag{5.6}$$

I then take the ratio of actual TFP to this efficient level of TFP to measure the counterfactual gains from reducing misallocation:

$$\frac{TFP_{js}}{TFP_{js,EFF}} = \left[\sum_i \left(\frac{A_{is}}{TFP_{js,EFF}} \frac{\overline{TFPR_{js}}}{TFPR_{ijs}}\right)^{\frac{1}{1-\eta}}\right]^{1-\eta}, \tag{5.7}$$

and aggregate it using a Cobb-Douglas aggregator across sectors within the prefecture:

$$\frac{TFP_j}{TFP_{j,EFF}} = \prod_s \sum_i \left(\frac{A_{is}}{TFP_{js,EFF}} \frac{\overline{TFPR_{js}}}{TFPR_{ijs}}\right)^{\frac{1}{1-\eta}}]^{\theta_s(1-\eta)}, \tag{5.8}$$

The percent gains in the prefecture are defined as

$$Gains_j = 100\left(\frac{TFP_{j,EFF}}{TFP_j} - 1\right) \tag{5.9}$$

Table 5
Counterfactual TFP Gains.

Year	(1) 1998	(2) 1999	(3) 2000
Panel A: Equalizing TFPR within prefectures			
%	90.6	97.1	102.0
Panel B: Relative to 1998 prefecture-level city gains			
%	16.1	18.1	23.1

Notes: Panel A reports counterfactual gains from equalizing TFPR within prefectures and sectors. Panel B reports counterfactual gains from moving the dispersions in decentralized prefectures to the relative efficient level in centralized prefecture-level cities.

Eq. (5.9) offers a way to calculate the counterfactual gains when dispersions of TFPR were reduced. Table 5 provides the results of two counterfactual practices. Panel A of Table 5 assumes that marginal products are equalized across all firms within each prefecture and sector. By this calculation, aggregate TFP in prefectures before they were treated can be raised by 90.6% in 1998 and 102.0% in 2000 without misallocation within each prefecture. These estimated gains are not easy to realize, as it is difficult to believe that the city government can get rid of all misallocation that exists in the original prefecture. To calculate a more realistic counterfactual gain, I take a conservative stance to hypothetically suppose the dispersions in the decentralized prefectures equal to the dispersion level in centralized cities, instead of the fully efficient level as above. These results are listed in Panel B of Table 5. Such counterfactual allocation will lead to a gain of 16.1% in TFP in 1998 and 23.1% in 2000.

Collectively, all results in this subsection paint a consistent picture. The centralization reform can reduce resource misallocation due to better coordination and reallocation by city governments. This improvement has a sizeable effect on aggregate productivity. It is noteworthy that these results using the firm-level dataset cover the period from 1998 to 2003. As introduced in the data section, over one third of prefectures received the centralization reform from 1998 to 2003, so the variations in this period are not trivial. In addition, the baseline results are also robust for the sample consisting only of these prefectures.

5.4. Decomposing the productivity gains

The results above find that the centralization reform helped to reduce resource misallocation and improve aggregate industrial productivity within cities. There are two ways to realize such gains in aggregate industrial productivity. First, incumbent firms can respond to the policy and adjust inputs and outputs (intensive margin); second, new firms with higher productivity can enter and incumbent firms with lower productivity can exit (extensive margin). In this subsection, I decompose estimated gains in aggregate productivity to intensive and extensive margins.

I follow Lu et al. (2019) to adopt the decomposition method proposed by Foster et al. (2001). Let A_{ijt} denote the TFP of the firm i in prefecture j and year t, and let x_{ijt} denote i's share of total employment in j. Changes in the aggregate TFP in prefecture j can be decomposed as follows:

$$\begin{aligned}\Delta TFP_{jt} = &\sum_{i\in C} x_{ijt-1}\Delta A_{ijt} + \sum_{i\in C}\left(A_{ijt-1} - TFP_{jt-1}\right)\Delta x_{ijt-1} + \sum_{i\in C}\Delta A_{ijt}\Delta x_{ijt} \\ &+ \sum_{i\in N} x_{ijt}\left(A_{ijt} - TFP_{jt-1}\right) - \sum_{i\in N} x_{ijt-1}\left(A_{ijt-1} - TFP_{jt-1}\right)\end{aligned} \tag{5.10}$$

where C denotes continuing firms, N denotes entering firms and X denotes exiting firms. The first three terms respectively represent within, between and cross-firm components based on firm-level changes in TFP and share. These gains are all contributed to by continuing incumbents, hence intensive margins. The last two terms represent the contribution of entering and exiting firms, hence extensive margins.

After the decomposing, I put each term into the baseline model and regress them on the treatment dummy. Table 6 shows the results.

Table 6
Decomposition: extensive and intensive margins.

VARIABLES	(1) Total	(2) Within	(3) Between	(4) Cross	(5) Entry-exit
Treat	0.052**	0.002	0.006	−0.009	0.052***
	(0.019)	(0.013)	(0.008)	(0.015)	(0.017)
Prefecture FE	✓	✓	✓	✓	✓
Year × province FE	✓	✓	✓	✓	✓
Observations	800	800	800	800	800
R-squared	0.521	0.024	0.367	0.519	0.054

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variables are different decomposing TFP gains. The main independent variable is whether this prefecture is treated to a prefecture-level city. Prefecture fixed effects and year × province fixed effects are controlled. Standard errors are clustered on province level.

Columns (2) to (5) reveal how much each term accounts for the total TFP gains shown in column (1). It is clear that the last extensive margin column contributes the most. Therefore, the decomposition exercise indicates that the newly created cities mostly improve aggregate industrial efficiency by allocating promoting extensive margins.

5.5. Concentration in space

As one of the most intuitive consequences of efficiency-improving reallocation, I will formally test whether industrial production becomes more concentrated and specialized within cities after the reform. Industrial concentration in China is believed to be lowered by political decentralization. Young (2000) is the first to document the fact that provinces in China converge to produce similar goods when they saw powers decentralized from the central government. Poncet (2003) shows that local protectionism under decentralization slowed down industrial concentration and agglomeration after 1980. Bai et al. (2004) find that geographic distributions of industries with high margins are more dispersed and appear to a higher extent due to local protectionism. Therefore, a natural implication of the centralization reform is that with better coordination, a specific industrial sector may become more concentrated geographically. We can expect that after the centralized city governments are created, local protectionism from the county governments is mitigated. Thus, it is not necessary for every county to own every industrial sector. Hence, a specific sector will be concentrated in fewer places.

To examine the concentration of industrial production in space, I make use of the γ index developed by Ellison and Glaeser (1997) to measure the geographic concentration of a specific sector within a prefecture. In their paper, the authors construct a model-based index of the geographic concentration of economic activities, the γ index. Lu and Tao (2009) calculate the γ index in a nationwide study and document the general trend of concentration in China. Here, I revise the γ index to fit into my county-prefecture scenario as follows:

$$\gamma_{sj} = \frac{G_{sj} - (1 - \sum_{c \in j} x_c^2) H_{sj}}{(1 - \sum_{c \in j} x_c^2)(1 - H_{sj})} \tag{5.11}$$

where γ_{sj} is the Ellison-Glaeser index calculated for each sector-prefecture pair sj; $G_{sj} = \sum_{c \in j}(x_c - s_{si})^2$ is the spatial Gini coefficient, where c is any county in prefecture j, x_c is c's share of total employment or output of all industries in j, s_{si} is sector s's share of employment or output for region r in county c; and H_{sj} is the Herfindahl index of sector s. The greater the Ellison-Glaeser index, the higher the geographic concentration. It equals zero if all the firms randomly pick their location. In this part, I calculate the Ellison-Glaeser index γ_{sjt} for every combination of year, sector and prefecture/city, and regress them on the treatment variable $Treat_{jt}$ as in the baseline model:

$$\gamma_{sjt} = \beta Treat_{jt} + \phi w_{jpt} + \alpha_j + \delta_{pt} + \theta_{st} + \epsilon_{jst} \tag{5.12}$$

where γ_{sjt} is the Ellison-Glaeser index in prefecture j, sector s and year t; w_{jpt} are baseline controls; α_j is prefecture fix effects; δ_{pt} is year × province fixed effects; and θ_{st} is sector × province fixed effects. The results are given in Table 7.

Table 7
Geographic Concentration.

VARIABLES	(1) EG Index-output	(2) EG Index-employment
Treat	0.029***	0.042***
	(0.008)	(0.008)
Prefecture FE	✓	✓
Year × province FE	✓	✓
Year × sector FE	✓	✓
Observations	45,286	45,286
R-squared	0.019	0.337

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variables are the Ellison-Glaeser Index respectively measured in outputs and employment. The main independent variable is whether this prefecture is treated to a prefecture-level city. Prefecture fixed effects, year × sector fixed effects and year × province fixed effects are controlled. Standard errors are clustered on province level.

The first column shows the result of the Ellison-Glaeser Index measured in outputs and the second shows the result measured in employment. From the results we find that after cities were created, the extent of industrial concentration increased significantly. It suggests that under centralization, goods in a given sector tend to be produced in fewer counties than under decentralization. The findings in this section support another well-known observation in Young (2000): decentralization in China induced convergence in the composition of outputs.

5.6. Evidence on centralization

In this subsection, I will investigate the nature of the 1983 reform in order to confirm that the reform since 1983 brought more centralization at the prefecture level. It is necessary to show that after the reform prefecture-level city governments have more powers, funding and responsibilities than before. To understand the nature of the reform, I bring in the fiscal data from the Public Finance Statistics Yearbooks as introduced in Section 3.1. I run regressions using the baseline model and outcomes measuring the size and powers of various governments, including the log value of government employment, the log value of government administrative expenditure, the share of fiscal revenue in the entire province, and the share of fiscal expenditure in the entire province (Li et al., 2016), respectively on prefecture level and county level. We expect the 1983 reform to be associated with increases in the size and powers of prefecture-level governments at the expense of county-level governments. In addition to those variables measuring centralization, I also check the effects on the transfer payments received from upper-level governments to check whether the reformed cities experience more favoritism from the central and provincial governments. The results are listed in Table 8.

Looking across column (1) to column (4) in Panel A, when a prefecture government is transformed to a prefecture-level city government, it will, as expected, become a "bigger" government. The number of its government employees will increase by 11.7% and administrative expenditures will increase by 36.5%. Its responsibilities for fiscal revenue and expenditure also rise significantly. At the same time, the decreases in the size of county-level governments are small and not significant, which reflects the difficultly in laying off public-sector employees in China. The share of expenditure and revenue by city governments in the entire province also increases significantly by 3.0 percentage points and 1.4 percentage points, respectively. The magnitudes of these coefficients are sizeable as they account for 20.9% and 18.8% of the average shares of prefecture-level governments in the entire country. The corresponding shares by county governments decrease at the same time. These re-

Table 8
Evidence on Centralization.

VARIABLES	(1) ln(gov't employ.)	(2) ln(admin. exp)	(3) exp. share	(4) rev. share	(5) transfer share
Panel A: Prefecture government					
Treat	0.117***	0.365***	0.044***	0.025**	−0.021
	(0.045)	(0.032)	(0.010)	(0.011)	(0.025)
Prefecture FE	✓	✓	✓	✓	✓
Year × province FE	✓	✓	✓	✓	✓
Observations	1378	1617	1617	1615	1455
R-squared	0.889	0.948	0.834	0.834	0.805
Panel B: County government					
Treat	−0.006	−0.009	−0.006**	−0.005*	0.011**
	(0.009)	(0.008)	(0.002)	(0.003)	(0.006)
County FE	✓	✓	✓	✓	✓
Year × province FE	✓	✓	✓	✓	✓
Observations	11,095	12,986	12,991	13,026	12,743
R-squared	0.945	0.946	0.525	0.522	0.865

Notes: *** denotes significance at 1%, ** at 5%, and * at 10%. Outcomes variables are log(numbers of government employees), log(administrative expenditure), percentage of expenditure in total expenditure of the province, percentage of revenue in total revenue of the province and percentage of transfer payment in total revenue. The main independent variable is whether this prefecture is treated to a prefecture-level city. Prefecture fixed effects and year × province fixed effects are controlled. Standard errors are clustered on province level.

sults support the centralization process and lay a solid basis of the whole story. In column (5), we find that transfer payment shares in total revenue of the city governments are not higher than those of the prefecture governments. This helps in excluding an alternative story whereby the treated regions become better simply because they receive more resources from upper-level governments. Nevertheless, this result cannot rule out all the other possible mechanism. For example, the central government may allocate more productive SOEs to the newly formed city, which cannot be observed in the fiscal data.

Admittedly, counties retain some fiscal powers after the reform. For example, some may be concerned that county governments exercise control over land, and land revenue is a type of extra-budgetary revenue that is not included in the above regressions. In fact, the land revenue and even the total extra-budgetary revenue only accounted for a small fraction in the total fiscal revenue before 2004 (Chen and Kung, 2016). The concern of the extra-budgetary revenue should not become a challenge during the period examined in this paper. To sum up, the results above increase our faith that the centralization reform raises the efficiency inside each boundary, and they are not simply receiving more help from outside.

6. Conclusion

In this paper, I investigate political centralization as a policy tool to help more productive urban areas through a reform in China to create cities. Both theoretical predictions and empirical evidence are ambiguous as to whether centralization or decentralization is better (Mookherjee, 2015). For example, the literature on fiscal federalism finds that decentralization from central to provincial governments can provide incentives to protect markets, and such findings have been applied fruitfully to Latin America and transition economies (Qian and Weingast, 1997). However, problems such as mismanagement, elite capture and corruption in local governments are found in studies on India and Africa (Bardhan, 2002). In the context of China, Young (2000) criticizes decentralization, as it induces local protectionism and a distortion of production, suggested by convergence in the composition of production and divergences in price and productivity across provinces. Consistent with Young (2000), this paper finds that decentralization distorts production and induces misallocation; creating cities to centralize county powers can in turn reallocate resources better and increase efficiency. Finally, centralization shapes more productive cities.

In this reform starting from 1983, China created prefecture-level cities in the basis of prefecture; this made the component counties less autonomous, and decision powers at the county level were centralized to city governments. After the reform, the city governments were responsible for economic policies and development in their component counties. Using a DID model and city-level data, I show that this centralization reform had positive causal effects on measures on urban industrial productivity. I use micro data on industrial firms and find the potential gains in productivity through reallocating resource and equalizing the marginal products of inputs across counties. Drawing on the misallocation methodology developed by Hsieh and Klenow (2009), I quantify such gains through a counterfactual exercise. The gains in aggregate productivity are mostly due to extensive margins involving in entry and exit of firms. Naturally, the reform improves higher industrial concentration and specialization in space, resulting a core-periphery structure after the reform (Krugman, 1991). Patterns in fiscal behavior of local governments confirm the existence of centralization and the reallocation of resources.

My findings have important policy implications for urbanization and industrialization in developing countries. First, it is accepted that the effects of decentralization differ case by case and through specific contexts, and designs matter considerably. This paper raises the potential value of centralization when we design policies in developing countries. In a context where local governments or communities have the power and incentives to distort resource allocation, this paper shows the inefficiencies that may arise from decentralization. It is therefore in line with Young (2000)'s observations. I show that in this setting centralization to a regional level mitigates misallocation problems with better coordination. Nevertheless, this paper will not challenge decentralization as a benchmark or argue that popular decentralization experiments should be abandoned. The main implication is that decentralization should not become an unconditionally promoted tool regardless of contexts and then require more elaborate policy designs, which is consistent with the suggestion by World Bank (2000).

Second, in classical works, the most popular reason for preferring decentralization is that it ensures better accountability from local governments since they compete with each other, while centralization may be problematic if central governments act as monopolists and lack accountability. However, in the setting of the present paper, centralization affected regions (prefectures/cities) rather than the central government. City government leaders also had to compete for promotion as county

leaders. As a result, in this arrangement, centralization at a regional level can avoid traditional problems in the political economy while still delivering benefits from better coordination. Admittedly, this setting would not be a first-best arrangement compared with the perfect decentralized equilibrium in Tiebout (1956). Nevertheless, in a real world with imperfect markets, distortion powers and incentives of local governments, this paper suggests centralization at a regional level to be a second-best choice.

Third, in the context of China, industrialization in urban areas is critical to maintain the rapid growth of the last 40 years. Therefore, the central government designs a number of policies to promote industrialization and urbanization, including the 1983 reform to create prefecture-level cities discussed in this paper. However, cities in China are still regarded as undersized and underurbanized (Au and Henderson, 2006). Henderson (2009) raises two policy suggestions in this regard: to integrate the urban and rural sectors and unify markets for input factors and to design an urban policy structure that provides appropriate incentives for public officials. The findings in this paper offer a solid evidence in support of the two suggestions. Through the 1983 reform to reshape the hierarchy between cities and counties, the city governments have the right incentives to coordinate resource allocation better and avoid setting barriers for input factor flows. With the combination of compatible incentives of local governments and an integrated regional market, we observe less misallocation and more productive industrial production at the city level. A more efficient industrial sector could in turn foster urbanization and attract more resources into industrial production.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jue.2019.06.005.

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